

# A PRACTICAL GUIDE TO INSTRUMENTAL VARIABLES METHODS WITH HETEROGENEOUS TREATMENT EFFECTS\*

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Instrumental variables (IV) methods represent one of the most popular approaches to identifying and estimating causal effects in economics. Currie et al. (2020) establish that the popularity of IV methods in applied microeconomics has grown continuously since the 1980s, as indicated by the content of the National Bureau of Economic Research (NBER) working papers. Likewise, Goldsmith-Pinkham (2026) documents that roughly 30% of NBER working papers in applied microeconomics mention instrumental variables.

Standard textbook treatments of IV methods assume a linear model with constant effects. At the same time, the influential local average treatment effect (LATE) framework of Imbens and Angrist (1994), Angrist and Imbens (1995), and Angrist et al. (1996) allows for a very general form of treatment effect heterogeneity, which the textbook model rules out. In this paper, we will offer a nontechnical, practical guide to the literature on IV methods with heterogeneous treatment effects, which originates from the work of Angrist and Imbens. Instead of aiming to provide a comprehensive survey of the literature, our focus will be on highlighting several areas in which existing work in applied microeconomics deviates from what we regard as best practices motivated by the recent theoretical literature.

The first area we explore is the choice of the target parameter. Recent research highlights possible differences between the local average treatment effect (LATE), as defined by Imbens and Angrist (1994), and the probability limits of usual IV and two-stage least squares (2SLS) estimators, especially in cases where the instrument is only valid conditional on covariates. As an alternative, we will discuss an intuitive, general strategy to estimate the LATE when

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covariates matter. The second area we consider is the flexibility of parametric specifications. Recent research shows that a causal interpretation of IV and 2SLS estimands requires that the conditional mean of the instrument given covariates be linear. Parametric misspecification is also possible when estimating the LATE with covariates. As a solution to such problems, we will discuss flexible estimation strategies based on machine learning. The third area we explore is possible violations of the assumptions underlying the LATE framework. These assumptions have testable implications that have spurred a sizable theoretical literature, yet have had limited influence on applied work. We will discuss the intuition behind the resulting tests and their implementation. We will also discuss estimation approaches that are more robust to violations of monotonicity, which is often a controversial assumption.

Throughout the paper, we focus on the standard LATE framework with a binary treatment and a binary instrument. Interested readers should also consider several other papers that offer a comprehensive survey of the existing literature, or focus instead on a subset of possible applications of IV methods. Mogstad and Torgovitsky (2018) use the marginal treatment effect (MTE) framework of Heckman and Vytlacil (2005) to discuss IV identification and extrapolation under treatment effect heterogeneity; the MTE is especially relevant with continuous instruments and can be understood as the LATE for a small change in the instrument. Mogstad and Torgovitsky (2024) review the literature on IV methods with heterogeneous treatment effects, including extensions to multivalued treatments, multivalued instruments, and multiple instruments. Borusyak et al. (2025) provide a guide to the literature on shift-share instruments. Chyn et al. (2025) and Goldsmith-Pinkham et al. (2025) survey the literature on judge leniency designs, an important IV application involving quasi-random assignment to many decision-makers rather than a single binary instrument.

## Target Parameters

In this section we discuss the differences between three leading estimands in applications of IV methods: the probability limit in the usual linear IV regression, the 2SLS estimand in a saturated specification with multiple interacted instruments, and the “true” local average treatment effect. We argue that applied researchers should more often estimate this last parameter, ideally as the main object of interest or at least as a robustness check. We also illustrate the differences between these parameters in two empirical applications.

### Motivating Example: Stratified RCTs with Imperfect Compliance

We start by considering a leading application of instrumental variables methods: a randomized controlled trial (RCT) with imperfect compliance. Concretely, let  $Y_i$  denote the

outcome,  $D_i$  denote the binary treatment, and  $Z_i$  denote the randomized assignment to treatment. We refer to  $Z_i$  as the instrument and leverage its randomness to estimate the causal effect of  $D_i$  on  $Y_i$ . Let  $D_i(1)$  and  $D_i(0)$  denote the two potential treatments, i.e., the counterfactual treatment statuses that would be observed if a unit were assigned ( $Z_i = 1$ ) or not assigned ( $Z_i = 0$ ) to be treated. Following the usual terminology, we refer to units with  $D_i(1) > D_i(0)$  as compliers. Under the standard assumptions underlying the LATE framework—namely, independence of the instrument, exclusion restriction, relevance, and monotonicity—the usual linear IV regression identifies the LATE, i.e., the average treatment effect for compliers (Imbens and Angrist, 1994). This is the so-called Wald estimand:  $\beta = [\mathbb{E}(Y_i | Z_i = 1) - \mathbb{E}(Y_i | Z_i = 0)] / [\mathbb{E}(D_i | Z_i = 1) - \mathbb{E}(D_i | Z_i = 0)] = \mathbb{E}[Y_i(1) - Y_i(0) | D_i(1) > D_i(0)]$ , where  $Y_i(1)$  and  $Y_i(0)$  are the treated and untreated potential outcomes.

Treatment assignment, however, is rarely *completely* random. Consider a stratified RCT where assignment is random only conditional on a specific set of covariates,  $X_i$ . For example, in the Oregon Health Insurance Experiment (OHIE), a randomized lottery that gave low-income adults the opportunity to apply for Medicaid, it is necessary to control for household size and survey wave. Although Medicaid eligibility was, for practical purposes, randomly assigned within household-size strata, assignment probabilities in the survey sample vary across cells defined by household size *and* survey wave because of the survey design (Finkelstein et al., 2012). The independence assumption,  $(Y_i(1), Y_i(0), D_i(1), D_i(0)) \perp Z_i$ , no longer holds unconditionally because these covariates may also be related to potential outcomes and treatments. Specifically, household size is plausibly related to health expenditures, health care utilization, and baseline health status, while survey wave may pick up time-varying shocks to outcomes. Consequently,  $Z_i$  is no longer independent of potential outcomes and treatments unless we control for these particular covariates.

There is not, however, a unique method to control for covariates, and this makes the LATE framework with covariates substantially more complicated than the baseline model without covariates. In what follows, we discuss three strategies to control for covariates in stratified RCTs with imperfect compliance, which we also summarize in Table 1. The first two are variations of the classical IV method that estimates a population outcome equation where covariates,  $X_i$ , are controlled in an additively separable fashion:

$$Y_i = \gamma'X_i + \beta D_i + e_i. \quad (1)$$

The difference is that the first approach, the *linear IV regression*, also uses an additively separable first stage,  $D_i = \delta'X_i + \pi Z_i + v_i$ . This approach yields a population coefficient  $\beta_{IV}$  on  $D_i$ . In the case of the OHIE, the elements of  $X_i$  are discrete and consist of mutually

exclusive indicators for different combinations of values of household size and survey wave. The method of analysis in Finkelstein et al. (2012) is equivalent to estimating  $\beta_{IV}$  using these covariates. To simplify the comparison with other approaches, we assume, for now, that  $X_i = (X_{i1}, \dots, X_{iJ})$  does indeed saturate the model. That is,  $X_i$  is a vector of dimension  $J$ , each entry  $X_{ij}$  is a dummy variable, and for every observation  $i$ ,  $\sum_j X_{ij} = 1$ .

The second approach, the *2SLS regression*, uses the same outcome equation in (1), but modifies the first stage by interacting covariates with the instrument:  $D_i = \delta' X_i + \sum_j \pi_j Z_i X_{ij} + v_i$ . Angrist and Pischke (2009) refer to this specification as the “saturate and weight” approach. Following Słoczyński (2026), who calls it “AI’s specification” after Angrist and Imbens (1995), we refer to the population coefficient on  $D_i$  as  $\beta_{AI}$ . With saturated covariates, the first-stage regression underlying  $\beta_{AI}$  is correctly specified by design, and each  $\pi_j$  may be interpreted as the causal effect of  $Z_i$  on  $D_i$  for the subgroup with covariate  $X_{ij} = 1$ , which is the same as the proportion of compliers in that subgroup.<sup>1</sup>

AI’s specification features a heterogeneous first stage, but retains the additively separable outcome equation in (1). To introduce the third approach, which we call the *LATE regression*, it is convenient to omit one element of the saturated vector  $X_i$ , include an intercept, and let  $\bar{X} = \mathbb{E}[X_i]$  denote the vector of covariate-cell proportions. Compared to AI’s specification, the LATE regression augments the outcome equation in (1) with interactions between the instrument and the centered covariates:

$$Y_i = \alpha + \beta_{LATE} \cdot D_i + \gamma' X_i + \eta' [Z_i \cdot (X_i - \bar{X})] + e_i, \quad (2)$$

which accommodates heterogeneity in both the first stage and the covariate-specific LATEs. Note that the first stage in AI’s specification can be reparameterized as  $D_i = \delta + \pi Z_i + \kappa' X_i + \phi' [Z_i \cdot (X_i - \bar{X})] + v_i$  without changing its fitted values. The LATE regression is therefore a just-identified IV regression in which 1,  $X_i$ , and interactions  $Z_i \cdot (X_i - \bar{X})$  are included as exogenous regressors, while  $Z_i$  is the instrument for  $D_i$ . By indirect least squares, the resulting estimator of  $\beta_{LATE}$  is equivalent to the ratio of two regression adjustment (RA) estimators: the OLS estimator of the coefficient on  $Z_i$  from a regression of  $Y_i$  on 1,  $Z_i$ ,  $X_i$ , and interactions  $Z_i \cdot (X_i - \bar{X})$ , divided by the estimated coefficient on  $Z_i$  from the corresponding regression with  $D_i$  as the dependent variable.<sup>2</sup> This representation mirrors the nonparametric identification result in Frölich (2007), according to which the LATE is identified as  $\mathbb{E}\{\mathbb{E}(Y_i | Z_i = 1, X_i) - \mathbb{E}(Y_i | Z_i = 0, X_i)\} / \mathbb{E}\{\mathbb{E}(D_i | Z_i = 1, X_i) - \mathbb{E}(D_i | Z_i = 0, X_i)\}$ , the

<sup>1</sup>We return to the issue of misspecification in the next section.

<sup>2</sup>Zhao et al. (2026, Section 6.3) briefly discuss this specification and note that the resulting coefficient on the treatment equals a ratio of regression-adjusted reduced-form and first-stage coefficients. They also contrast it with their “interacted 2SLS” specification, which instead includes interactions between the treatment and appropriately centered covariates in the outcome equation.

ratio of the average effects of the instrument on the outcome and on the treatment. This result relies on the LATE assumptions extended to settings with covariates (cf. Abadie, 2003), but does not impose constant treatment effects.

This nonparametric identification result holds even when  $X_i$  contains general, rather than discrete and saturated, covariates. Thus, the linear specification in (2) can be applied with general  $X_i$ , providing a starting point for approximating the conditional mean functions entering the numerator and denominator. Alternatively, one may instead use inverse probability weighting (IPW) or doubly robust (DR) methods to estimate the numerator and denominator in Frölich’s formula. With saturated discrete covariates, standard implementations of these approaches are numerically equivalent and reduce to computing the effect of the instrument on the outcome and treatment within each covariate cell, using differences in means, and then aggregating the cell-specific effects using the proportions of observations in the respective cells. When  $X_i$  is not saturated, however, these approaches need not coincide because they rely on different approximations for the relevant conditional means and instrument propensity scores, which we return to below.

When treatment effects are heterogeneous,  $\beta_{IV}$ ,  $\beta_{AI}$ , and  $\beta_{LATE}$  are not the same; each parameter corresponds to a different object with its specific interpretation. To see this, under the same assumptions as above (and with saturated covariates for simplicity), we can derive a weighted average representation of the three estimands in terms of covariate-specific LATEs, that is, the average treatment effects for compliers with covariate level  $j$ ,  $\tau_j = \mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0), X_{ij} = 1]$ . Indeed, all three estimands can be written as  $\beta_m = \sum_j \frac{\omega_{m,j}}{\sum_k \omega_{m,k}} \tau_j = \sum_j w_{m,j} \cdot \tau_j$ , where the estimand-specific unnormalized weights,  $\omega_{m,j}$ , are given in Table 1. The representations of  $\beta_{IV}$  and  $\beta_{AI}$  follow from Słoczyński (2026); the representation of  $\beta_{AI}$  is also implied by Theorem 3 of Angrist and Imbens (1995). The representation of  $\beta_{LATE}$  follows from Frölich (2007).

The expression for  $\beta_{LATE}$  is the most intuitive, aggregating  $\tau_j$  based on the probability of compliance at each covariate level, which implies that  $\beta_{LATE}$  is equal to the overall average treatment effect for compliers,  $\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)]$ . In other words, each (normalized) weight is the proportion of all compliers in the corresponding covariate cell. On the other hand,  $\beta_{IV}$  and  $\beta_{AI}$  introduce additional terms to their respective weights, which complicates interpretation. Specifically,  $\beta_{IV}$  overweights the cells with large values of  $\text{Var}[Z_i \mid X_{ij} = 1]$ , while  $\beta_{AI}$  overweights those with large values of the product of  $\pi_j$  and  $\text{Var}[Z_i \mid X_{ij} = 1]$  (Słoczyński, 2026). In a model with homogeneous treatment effects in which (1) is correctly specified, such weighting may be desirable because it improves estimation efficiency; with heterogeneous treatment effects, however, it would be difficult to justify an explicit interest in  $\beta_{IV}$  or  $\beta_{AI}$ . Ultimately, the average treatment effect for

compliers,  $\beta_{\text{LATE}}$ , may differ substantially from  $\beta_{\text{IV}}$ ,  $\beta_{\text{AI}}$ , or both. Such differences can arise when  $\pi_j$ , the conditional proportion of compliers, or  $\text{Var}[Z_i | X_{ij} = 1]$ , the conditional variance of the instrument, vary across covariate values.<sup>3</sup> Whether these differences are large in a given application, however, is an empirical question.

To illustrate this empirical question, we use data from Finkelstein et al. (2012). In this setting, as mentioned above, the instrument (the ability to apply for Medicaid) is randomized conditional on household size and survey wave, and the identifying assumptions underlying the LATE framework seem plausible. In our subsequent analysis, we focus on the impact of Medicaid coverage on the number of outpatient visits in the last six months. To simplify the illustration, unlike Finkelstein et al. (2012), we do not use survey weights.

Panel A of Table 2 reports the estimates of  $\beta_{\text{LATE}}$ ,  $\beta_{\text{IV}}$ , and  $\beta_{\text{AI}}$  using the full sample from the OHIE. Although there is substantial heterogeneity in covariate-specific LATEs, the different estimands yield nearly identical point estimates and standard errors, suggestive of a precisely estimated increase in the number of outpatient visits in response to Medicaid enrollment by about 1 in six months on average. Apparently, the correlation between  $\widehat{\text{Var}}[Z_i | X_{ij} = 1]$  and  $\hat{\tau}_j$  as well as  $\hat{\pi}_j \cdot \widehat{\text{Var}}[Z_i | X_{ij} = 1]$  and  $\hat{\tau}_j$  is sufficiently weak among compliers to render the differences between  $\hat{\beta}_{\text{LATE}}$ ,  $\hat{\beta}_{\text{IV}}$ , and  $\hat{\beta}_{\text{AI}}$  meaningless.

The picture is very different, however, when we restrict our attention to the subsample of three-person households in Panel B of Table 2. This restriction does not alter the identifying assumptions, i.e., randomization continues to hold conditional on survey wave, but it changes the practical importance of the different components in the weighted average representations in Table 1. Indeed, the estimated LATE, 4.022, is now substantially larger than  $\hat{\beta}_{\text{IV}}$  and  $\hat{\beta}_{\text{AI}}$ , which are equal to 2.790 and 2.268, respectively.

The differences between  $\hat{\beta}_{\text{LATE}}$ ,  $\hat{\beta}_{\text{IV}}$ , and  $\hat{\beta}_{\text{AI}}$  follow directly from their distinct weighting schemes, summarized in Table 1. The estimated LATE aggregates covariate-specific treatment effects in proportion to each cell's estimated share among compliers,  $\hat{p}_j \hat{\pi}_j / \sum_k \hat{p}_k \hat{\pi}_k$ . By contrast, the IV estimate also weights each cell by the conditional variance of the instrument,  $\widehat{\text{Var}}[Z_i | X_{ij} = 1]$ , while the 2SLS estimate further weights each cell by  $\hat{\pi}_j$ . As a result, these estimates place relatively more weight on covariate cells with greater identifying variation, which need not be the cells that are most representative of the complier population.

Panel C of Table 2 illustrates this mechanism. For each of the three covariate cells in the subsample of three-person households, we report its sample proportion,  $\hat{p}_j$ , the conditional variance of  $Z_i$ , as well as the estimated proportion of compliers,  $\hat{\pi}_j$ , and conditional LATE,  $\hat{\tau}_j$ . We also report the estimated weights underlying each of the estimands, which are the normal-

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<sup>3</sup>The interpretation of  $\beta_{\text{LATE}}$ ,  $\beta_{\text{IV}}$ , and  $\beta_{\text{AI}}$  will change when the monotonicity assumption is violated. We return to this issue below.

ized sample counterparts of the expressions in Table 1. For example,  $\hat{w}_{\text{LATE},j} = \hat{p}_j \hat{\pi}_j / \sum_k \hat{p}_k \hat{\pi}_k$  and  $\hat{w}_{\text{IV},j} = \hat{p}_j \hat{\pi}_j \cdot \widehat{\text{Var}}[Z_i | X_{ij} = 1] / \sum_k \hat{p}_k \hat{\pi}_k \cdot \widehat{\text{Var}}[Z_i | X_{ik} = 1]$ . In the first covariate cell,  $\hat{w}_{\text{LATE},1} = (0.241 \cdot 0.455) / (0.241 \cdot 0.455 + 0.448 \cdot 0.280 + 0.310 \cdot 0.250) \approx 0.351$  and  $\hat{w}_{\text{IV},1} = (0.241 \cdot 0.168 \cdot 0.455) / (0.241 \cdot 0.168 \cdot 0.455 + 0.448 \cdot 0.037 \cdot 0.280 + 0.310 \cdot 0.099 \cdot 0.250) \approx 0.600$ . (All values are rounded for illustration.) Each of the estimates reported in Panel B of Table 2 can be obtained as the dot product of  $\hat{\tau}_j$  and the respective weights. For example,  $\hat{\beta}_{\text{LATE}} = 0.351 \cdot 1.067 + 0.401 \cdot 6 + 0.248 \cdot 5 \approx 4.022$  and  $\hat{\beta}_{\text{IV}} = 0.600 \cdot 1.067 + 0.151 \cdot 6 + 0.249 \cdot 5 \approx 2.790$ . In this subsample, because  $\widehat{\text{Var}}[Z_i | X_{ij} = 1]$  and  $\hat{\pi}_j$  are large when  $\hat{\tau}_j$  is small,  $\hat{\beta}_{\text{IV}}$  and  $\hat{\beta}_{\text{AI}}$  underweight the covariate cells with large estimated treatment effects, producing a downward bias relative to the LATE. As a result, the three estimands answer different causal questions despite relying on the same experimental variation.<sup>4</sup> A cautious researcher should investigate whether a similar pattern arises in their specific setting.

## Beyond Stratified RCTs

In the stratified RCT setting above, strata indicators saturate the relevant covariate space by construction. In other IV applications, however, saturated specifications may be infeasible, especially when covariates are numerous or continuously distributed. The nonparametric identification result of Frölich (2007) continues to apply in such settings. By contrast, interpreting the usual linear IV estimand,  $\beta_{\text{IV}}$ , as a weighted average of causal effects requires an additional parametric restriction: the conditional mean of  $Z_i$  given  $X_i$  must be linear in  $X_i$ . Blandhol et al. (2026) call this the “rich covariates” assumption and show that it is necessary for a causal interpretation of  $\beta_{\text{IV}}$ .

As in stratified RCTs, we recommend reporting an estimate of  $\beta_{\text{LATE}}$  when covariates matter but saturated specifications are infeasible, even if only as a robustness check alongside  $\hat{\beta}_{\text{IV}}$ . The two parameters,  $\beta_{\text{IV}}$  and  $\beta_{\text{LATE}}$ , may often be similar, but when they differ,  $\beta_{\text{LATE}}$  is arguably of greater substantive interest. (Because  $\beta_{\text{AI}}$  is also unlikely to be the main object of interest when it differs from  $\beta_{\text{LATE}}$ , we focus on the comparison between  $\beta_{\text{IV}}$  and  $\beta_{\text{LATE}}$  for simplicity.) As discussed above, the result in Frölich (2007) implies that  $\beta_{\text{LATE}}$  can be estimated as the ratio of the average effect of the instrument on the outcome to its average effect on the treatment. Without saturation, these two average effects can be estimated in

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<sup>4</sup>One way to see this is through the lens of the framework developed by Poirier and Słoczyński (2025) to quantify the internal validity and representativeness of various estimands. If we treat the population of compliers—the largest subpopulation for which the average treatment effect is nonparametrically identified in an IV setting under standard assumptions—as our target, then  $\beta_{\text{LATE}}$  is associated with the largest possible value of the measure of internal validity, equal to 1, while the measures associated with  $\beta_{\text{IV}}$  and  $\beta_{\text{AI}}$  will generally be less than 1. This means that these estimands correspond to average treatment effects for some (possibly small) subsets of the complier subpopulation, whereas  $\beta_{\text{LATE}}$  is the average over all compliers.

a number of distinct ways, implying a corresponding range of suitable estimators of  $\beta_{\text{LATE}}$ .<sup>5</sup>

In principle, one could estimate the equation in (2) by instrumental variables and report the corresponding regression adjustment estimate of  $\beta_{\text{LATE}}$ . We instead focus on an inverse probability weighting (IPW) estimator of the same parameter, both to make the contrast with  $\hat{\beta}_{\text{IV}}$  clearer and to set up the discussion of the “rich covariates” assumption below. The IPW estimator is based on a first-step estimate of the conditional mean of  $Z_i$  given  $X_i$ , often called the instrument propensity score. This is the same conditional mean that is central to the “rich covariates” assumption. Here, however, the goal is to estimate  $\beta_{\text{LATE}}$ , and one would typically use a logit or probit model rather than the linear probability model implicit in  $\beta_{\text{IV}}$ . The resulting instrument propensity score estimates are used to reweight units with  $Z_i = 1$  and  $Z_i = 0$  so that the two subsamples are comparable in terms of their covariates. This enables straightforward estimation of the average effects of the instrument on the outcome and on the treatment using differences in reweighted means. The ratio of these two differences is the IPW estimator proposed by Frölich (2007) for the LATE. Słoczyński et al. (2025) emphasize the importance of using normalized weights in this context, explore the connection to the “kappa weighting” of Abadie (2003), and develop the Stata package `kappalate`. The package allows binary and nonbinary treatments, whether discrete or continuous; the instrument, however, must be binary.

To illustrate the possible differences between  $\beta_{\text{IV}}$  and  $\beta_{\text{LATE}}$  in a nonsaturated setting, we estimate both parameters using Abadie’s (2003) sample from the 1991 Survey of Income and Program Participation (SIPP). In this application, based on Poterba et al. (1994), one of the causal effects of interest is that of participation in a 401(k) retirement plan on participation in an individual retirement account (IRA). While 401(k) participation is likely endogenous, Poterba et al. (1994) and Abadie (2003) argue that 401(k) eligibility, which is determined by the employer, can be used as an instrument for 401(k) participation. Because 401(k) eligibility is not randomized or “as good as randomly assigned,” Abadie (2003) controls for family income, age, age squared, marital status, and family size to estimate the causal effects of 401(k) participation using this instrument.

Panel A of Table 3 replicates the linear IV estimate of the effect of 401(k) on IRA participation, as reported in column (2) of Table 3 in Abadie (2003). This estimate, statistically significant at the 5% level, suggests that participation in a 401(k) retirement plan increases the probability of IRA participation by about 2.74 percentage points. However, Panel A of Table 3 also reports two IPW estimates of the LATE. These estimates, based on logit

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<sup>5</sup>Some of these estimators include those in Tan (2006), Frölich (2007), Uysal (2011), Heiler (2022), Sant’Anna et al. (2022), Słoczyński et al. (2022, 2025), Ma et al. (2026), and Zhao et al. (2026). Related machine learning approaches have been developed by Belloni et al. (2017), Chernozhukov et al. (2018), Sun and Tan (2022), Singh and Sun (2024), and others.



and probit instrument propensity scores, are much smaller than the linear IV estimate, with  $p$ -values equal to 0.221 and 0.197, respectively. This calls into question the conclusion about the positive impact of 401(k) on IRA participation. If we ignore estimation uncertainty and the possibility of misspecification, the difference between  $\hat{\beta}_{IV}$  and  $\hat{\beta}_{LATE}$  must be driven by the different weights underlying the corresponding estimands. In our view, the possibility that such differences may materialize necessitates explicit estimation of  $\beta_{LATE}$  in relevant applied work—either as the main parameter of interest or at least as a robustness check.

Panel A of Appendix Table A1 lists several R and Stata packages, including `kappalate`, which can be used to estimate the LATE when covariates matter.

## Parametric Misspecification

In this section we discuss the possibility that estimates of  $\beta_{IV}$  and  $\beta_{LATE}$  may be biased due to parametric misspecification. If all the requisite covariates are observed and controlled for, the leading case of misspecification occurs when important interaction and higher-order terms are omitted. First, we discuss how this concern applies to estimation of both  $\beta_{IV}$  and  $\beta_{LATE}$ , and how it is possible to test for parametric misspecification. Second, we recommend modern estimation approaches based on machine learning as a general solution to this problem.

### Rich Covariates

A large literature on weighted average representations of  $\beta_{IV}$ ,  $\beta_{AI}$ , and related estimands has used the assumption that the conditional mean of  $Z_i$  given  $X_i$  is linear in  $X_i$ . As mentioned in the previous section, a recent paper by Blandhol et al. (2026) shows that this assumption, termed “rich covariates,” is not only sufficient but also *necessary* for the resulting estimand to represent a nonnegatively weighted average of complier causal effects. If covariates are not rich, the estimand is not “weakly causal,” which means that when all conditional average treatment effects have the same sign, the estimand may assume the opposite sign. Poirier and Słoczyński (2025) show that this is equivalent to the absence of any subpopulation of compliers whose average treatment effect is recovered by the estimand of interest regardless of the pattern of treatment effect heterogeneity.

An immediate implication is that applied researchers targeting simple instrumental variables estimands should only use covariate specifications that are plausibly rich. To examine whether a given specification is rich, Blandhol et al. (2026) recommend the regression specification error test (RESET) of Ramsey (1969). In its canonical form, RESET, as applied to this problem, consists of estimating a regression of  $Z_i$  on  $X_i$  using OLS, obtaining fitted values,  $\hat{Z}_i$ , and, finally, estimating a regression of  $Z_i$  on  $X_i$  and several higher-order terms

of  $\hat{Z}_i$ , such as  $\hat{Z}_i^2$ ,  $\hat{Z}_i^3$ , and  $\hat{Z}_i^4$  (again using OLS). In this setup, RESET is simply the  $F$  test of joint significance of the higher-order terms in the second-step regression, and should be interpreted as a test for neglected nonlinearity (Wooldridge, 2010, Section 6.3.3). If we reject the null hypothesis, we conclude that the original covariate specification is not rich.

It would be inaccurate to suggest that estimators of  $\beta_{\text{LATE}}$  cannot suffer from similar specification problems. For example, the IPW estimator discussed in the previous section relies on the assumption that the conditional mean of  $Z_i$  given  $X_i$  is correctly specified, *however it is specified*. This offers more flexibility than the result in Blandhol et al. (2026), because we are no longer tied to the linear probability model and can instead use, say, the logit or probit model. On the other hand, the logit or probit specification will not be correctly specified if we fail to include relevant interaction and higher-order terms, as in the case of “rich covariates.” To test for parametric misspecification of the logit or probit model for the instrument propensity score, one may use the extension of RESET proposed by Papke and Wooldridge (1996). Here, we need to augment the original specification with several higher-order terms of the fitted linear index and, again, test for their joint significance.

Table 3 illustrates the problem of parametric misspecification. In addition to the previously discussed estimates of  $\beta_{\text{IV}}$  and  $\beta_{\text{LATE}}$ , Panel A reports the RESET  $p$ -values associated with each. With the original covariate specification, we decidedly reject the “rich covariates” assumption but also the correct specification of the logit and probit models for the instrument propensity score. Can a more flexible specification salvage a causal interpretation of our estimates? Panel B uses a quintic in family income, a quintic in age, an indicator for marital status, and indicators for each value of family size. With this set of covariates, RESET does not reject any of the null hypotheses of correct specification of the linear probability, logit, and probit models.  $\hat{\beta}_{\text{IV}}$  and  $\hat{\beta}_{\text{LATE}}$  are now very similar to each other, although  $\hat{\beta}_{\text{LATE}}$  has barely changed relative to Panel A while  $\hat{\beta}_{\text{IV}}$  has decreased by about 40%.

This illustration raises the question of whether  $\hat{\beta}_{\text{IV}}$  is less robust to parametric misspecification than estimators of  $\beta_{\text{LATE}}$ . Although we are not aware of such results for generic IPW estimators, other approaches to estimating  $\beta_{\text{LATE}}$  offer improved robustness properties, including the covariate balancing estimators of Heiler (2022), Sant’Anna et al. (2022), and Słoczyński et al. (2025), and the doubly robust estimators of Tan (2006), Uysal (2011), Słoczyński et al. (2022), and Ma et al. (2026). Several of these estimators are implemented in the Stata package `drlate` (Słoczyński et al., 2022), which also accommodates models for count outcomes with many zeros, such as the number of outpatient visits in Table 2. We recommend wider use of these covariate balancing and doubly robust estimators, together with their machine learning counterparts discussed below.

## Double Machine Learning

The previous subsection explained that a “weakly causal” interpretation of  $\beta_{IV}$  and  $\beta_{LATE}$  may fail when the conditional mean of the instrument given covariates is misspecified, perhaps due to omitted interaction or higher-order terms. One practical response is to manually enrich the covariate specification and then apply RESET. In many empirical applications, however, the number of potential interactions and nonlinear transformations is large, making manual specification and specification testing cumbersome.

A complementary approach is to use double/debiased machine learning (DML) to flexibly approximate the relevant conditional means. DML methods for estimating  $\beta_{LATE}$  have been developed by Belloni et al. (2017), Chernozhukov et al. (2018), Sun and Tan (2022), and Singh and Sun (2024), whereas Chernozhukov et al. (2018) also applied the DML methodology to estimating  $\beta_{IV}$ . An application of double/debiased machine learning in an IV context proceeds as follows. The researcher supplies a rich dictionary of covariates to a machine learning algorithm (e.g., lasso, ridge, or regression tree). This method then approximates the relevant nonlinear relationships—the conditional means of potential outcomes and potential treatments, as well as that of the instrument—in a data-driven way, with tuning parameters typically chosen by cross-validation. Finally, DML combines the resulting flexible predictions with orthogonal scores and cross-fitting to decrease the underlying regularization bias. In this sense, DML can be viewed as an automated approach to constructing “rich” covariate specifications, reducing the need for ad hoc choices of interaction and higher-order terms.

Relative to manually specified parametric models, restrictions of DML are less explicit and arise implicitly from the choice of the machine learning algorithm. For example, lasso assumes that the target functions can be well approximated by a sparse subset of the dictionary, while tree-based methods rely on recursive partitioning and local averaging. Thus, even in the DML framework, the validity of the researcher’s conclusions continues to hinge on the chosen approximation. Accordingly, it is still recommended to compare multiple machine learning algorithms to assess robustness and to reduce reliance on the approximation properties, and thus the implicit parametric restrictions, of any single method.

Table 4 illustrates these considerations by reporting DML estimates of  $\beta_{IV}$  and  $\beta_{LATE}$  in the 401(k) application considered above. In the case of both parameters, we estimate the relevant conditional means using six different machine learning algorithms: lasso, elastic net, ridge, random forest, regression tree, and XGBoost. The resulting estimates are generally smaller than the initial linear IV estimate in Panel A of Table 3 but larger than the corresponding LATE estimates. At the same time, the DML estimates remain somewhat sensitive to the choice of the machine learning algorithm. Thus, while DML relaxes explicit functional form assumptions, it is still advisable to verify robustness across alternative approximation

methods (as we do in Table 4). Because no single machine learning algorithm performs best in all settings, we recommend that researchers routinely report estimates from at least two or three algorithms to assess the sensitivity of their conclusions.

As an alternative to reporting multiple estimates separately, researchers can use stacking, a model averaging approach that combines several algorithms into a single estimate. Table 4 also reports the short-stacking estimate recommended by Ahrens et al. (2025). This estimate combines all six algorithms by assigning them nonnegative weights that sum to one. Rather than relying on any single learner, stacking lets the data determine the optimal combination of algorithms and therefore provides a useful summary of the evidence across methods. In our application, the short-stacking estimates are broadly consistent with the individual learner estimates, suggesting that our conclusions are not sensitive to the choice of algorithm.

Panel B of Appendix Table A1 lists the leading R and Stata packages that can be used to implement DML methods.

## Assumption Violations

In this section we advise practitioners to systematically assess whether the assumptions underlying the LATE framework may be violated. We begin by reviewing several testable implications of these assumptions, and explain how the resulting statistical tests can be implemented in practice. We argue that using such tests can give more credence to IV applications. We also discuss an intuitive estimation approach that is more robust to violations of monotonicity, which is a controversial assumption in many applications. The validity of this assumption may also be rejected by one of the statistical tests that we review, in which case such alternative estimation approaches are particularly useful.

### Testing Instrument Validity

As first observed by Balke and Pearl (1997), the standard LATE assumptions—specifically, independence of the instrument, exclusion restriction, and monotonicity—have implications that can be used to test instrument validity. If the LATE assumptions hold, then it must be the case that

$$\mathbb{P}(Y_i \in A, D_i = 1 \mid Z_i = 1) - \mathbb{P}(Y_i \in A, D_i = 1 \mid Z_i = 0) \geq 0 \quad (3a)$$

and

$$\mathbb{P}(Y_i \in A, D_i = 0 \mid Z_i = 0) - \mathbb{P}(Y_i \in A, D_i = 0 \mid Z_i = 1) \geq 0 \quad (3b)$$

for any subset  $A$  of outcome values (e.g., any interval or range). Because the inequalities in (3) are defined in terms of moments of observed variables, they can be examined with empirical data. Kitagawa (2015) derived a formal statistical test based on these inequalities, and proved that they are sharp (i.e., cannot be improved upon) as well as necessary but not sufficient for instrument validity. In other words, we may be able to reject invalid instruments, but it is fundamentally impossible to confirm instrument validity, even with infinite data. In a subsequent contribution, Huber and Mellace (2015) developed a sharp test of weaker assumptions that still suffice to identify the LATE. Mourifié and Wan (2017) reformulated the inequalities in (3) as two conditional moment inequalities, with  $Y_i$  entering as a conditioning variable, which facilitates implementation. Finally, Sun (2023) and Kwon and Roth (2026) extended this framework to the case of multivalued treatments.

A careful reader might ask: *Why* is it the case that the inequalities in (3) must hold whenever an instrument is valid? One answer to this question follows from Imbens and Rubin (1997), who proved that the difference in (3a) identifies the joint distribution of the treated outcome and complier status,  $\mathbb{P}(Y_i(1) \in A, D_i(1) > D_i(0))$ , while the difference in (3b) identifies  $\mathbb{P}(Y_i(0) \in A, D_i(1) > D_i(0))$ , the joint distribution of the untreated outcome and being a complier. Because these probabilities, like any other probabilities, must be nonnegative, the differences in (3) must be nonnegative, too.

Useful intuition for these testable implications is also provided by Kwon and Roth (2026). Note that the difference in (3a) is simply the coefficient on  $Z_i$  in the population regression of the compound outcome  $1[Y_i \in A, D_i = 1]$  on  $Z_i$ . If the instrument  $Z_i$  is valid and there are no defiers, any difference can only be driven by always-takers, never-takers, and compliers. In the case of always-takers and never-takers,  $Z_i$  has no effect on  $D_i$ . If so, then the exclusion restriction implies that it also has no effect on  $Y_i$ , ensuring that the effect of  $Z_i$  on the compound outcome is zero for both groups. It follows that the entire difference in (3a) is driven by compliers. However, for compliers,  $Z_i = 0$  implies that  $D_i = 0$ , which means that  $\mathbb{P}(Y_i \in A, D_i = 1 \mid Z_i = 1) - \mathbb{P}(Y_i \in A, D_i = 1 \mid Z_i = 0) = \mathbb{P}(Y_i \in A, D_i = 1 \mid Z_i = 1)$ . This probability, too, cannot be negative, which is exactly what the inequality in (3a) requires. A similar intuition justifies the inequality in (3b).

Although tests based on the implications of the LATE assumptions remain underused in applied work, their results should be interpreted with caution. As noted above, rejection refutes instrument validity, but a failure to reject should not be interpreted as evidence of validity, even in the population. This asymmetry arises because the testable implications are necessary, but not sufficient, for instrument validity. An additional limitation is that these tests are most likely to detect violations when the reduced form effects are sufficiently large relative to the first stage. When the first stage is strong, violations of the LATE assumptions

may therefore be harder to detect; in the limiting case of perfect compliance, the testable implications are automatically satisfied even if the exclusion restriction fails. Thus, a failure to reject remains especially uninformative in settings with strong first stages.

These tests may also be difficult to implement in some applications, especially when the instrument is valid only conditional on covariates. In such cases, the covariates enter the inequalities in (3) as conditioning variables, which increases the computational burden. To facilitate implementation, Farbmacher et al. (2022) proposed testing the Balke and Pearl conditions using machine learning.<sup>6</sup> Specifically, their procedure uses causal forests to estimate the conditional average effects of  $Z_i$  on the compound outcomes  $1[Y_i \in A, D_i = 1]$  and  $-1[Y_i \in A, D_i = 0]$ , and then identifies covariate groups with likely assumption violations. With sample splitting, group selection in one subsample still enables valid testing in another. Beyond the ease of implementation, tests based on machine learning make it possible to detect violations that occur only in some regions of the covariate space and would otherwise be masked by aggregation.

One limitation of the procedure in Farbmacher et al. (2022) is that it only considers the Balke and Pearl conditions, which require discretizing the outcome variable to construct the compound outcomes. This would not be necessary, for example, when testing the Mourifié and Wan conditions. It also seems reasonable to use a similar machine-learning-based approach to construct a formal test of the implication of independence and monotonicity that the conditional first stage,  $\mathbb{E}(D_i \mid Z_i = 1, X_i) - \mathbb{E}(D_i \mid Z_i = 0, X_i)$ , must be nonnegative at all covariate values.<sup>7</sup> These testable implications of the LATE assumptions, as well as other features, are implemented in the R package `montest` (Andresen, 2026).

Table 5 reports the resulting  $p$ -values for three applications: the Oregon Health Insurance Experiment, as in Table 2, the 401(k) application of Abadie (2003), as in Tables 3 and 4, and the study of causal effects of pretrial detention on case outcomes in Stevenson (2018). This last application uses data on more than 300,000 Philadelphia arrests between 2006 and 2013, and instruments for pretrial detention with indicators for randomly assigned judges. Following the replication of Stevenson (2018) in Słoczyński (2026), we use incarceration length as the outcome variable as well as a saturated covariate specification with indicators for each combination of offense type, race and gender of the defendant, and three time periods considered by Stevenson (2018). As in Słoczyński (2026), we also focus on a binary

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<sup>6</sup>As an alternative, Carr and Kitagawa (2023) and Mao et al. (2026) show how to incorporate covariates when testing the identifying assumptions in the MTE framework of Carneiro et al. (2011).

<sup>7</sup>This implication is sometimes tested by practitioners, especially in applications of the judge leniency design, but the choice of groups to examine is typically ad hoc. See, for example, Maestas et al. (2013), Dobbie et al. (2018), and Autor et al. (2019). An appropriate procedure based on machine learning can systematically search for groups where violations of monotonicity are most likely to be found.

instrument indicating whether a given case was assigned to the most lenient judge, although a more faithful replication of Stevenson (2018) would instead retain the eight original judge instruments; see, e.g., Mogstad and Torgovitsky (2024).

The  $p$ -values in Table 5 clearly indicate that there is fundamentally no evidence against instrument validity in the first two applications, regardless of whether we consider the Balke and Pearl (“BP”) conditions, the Mourifié and Wan (“MW”) conditions, or the nonnegativity of the conditional first stage (“FS”). In fact, this last condition is trivially satisfied in the 401(k) application of Abadie (2003), which is characterized by one-sided noncompliance. At the same time, we clearly reject the null hypothesis of instrument validity using the data from Stevenson (2018). The conclusion is essentially the same whether we consider the testable implications of independence, exclusion, and monotonicity (BP and MW) or the testable implications of independence and monotonicity (FS). Assuming that independence is satisfied, which is plausible, this implies that either monotonicity is violated but exclusion is satisfied, or both monotonicity and exclusion are violated. Our discussion in the next subsection will implicitly assume the former possibility.

Panel C of Appendix Table A1 lists several R packages, including `montest`, which can be used to implement various tests of the LATE assumptions.

## Robustness to Monotonicity Violations

If the monotonicity assumption is rejected or otherwise implausible, as in Stevenson (2018), estimation of  $\beta_{\text{LATE}}$  will be difficult, and the previously discussed estimators of  $\beta_{\text{LATE}}$  and  $\beta_{\text{IV}}$  will not consistently estimate any “weakly causal” target parameter. At the same time, appropriate estimators of  $\beta_{\text{AI}}$  will not be subject to such concerns, at least when a weaker assumption, termed “weak monotonicity” in Słoczyński (2026), is assumed to hold. Weak monotonicity, unlike strong monotonicity (i.e., the usual assumption), allows both compliers and defiers to exist, but they must not coexist at any value of covariates; that is, weak monotonicity requires that there be no defiers at some covariate values and no compliers elsewhere. This assumption will be plausible in some applications, but not in others. In Stevenson (2018), it seems reasonable that only a small number of observed characteristics of cases and defendants would determine the relative leniency of any particular judge.

To see why standard estimators of  $\beta_{\text{LATE}}$  and  $\beta_{\text{IV}}$  will not work under weak monotonicity, note that the term  $\pi_j$  in the weighted average representations in Table 1 only retains its interpretation as the conditional proportion of compliers under strong monotonicity. In the absence of any assumptions, it is simply equal to  $\mathbb{E}(D_i \mid Z_i = 1, X_i) - \mathbb{E}(D_i \mid Z_i = 0, X_i)$ , that is, the conditional first stage. (Under strong monotonicity, the conditional first stage identifies the conditional proportion of compliers.) However, under weak monotonicity, the

conditional first stage will be positive at some covariate values and negative at others, which directly translates to the incidence of “negative weights” in  $\beta_{\text{LATE}}$  and  $\beta_{\text{IV}}$  (as evident in Table 1). On the other hand,  $\pi_j$  enters the expression in  $\beta_{\text{AI}}$  as a quadratic rather than linearly, which ensures that the resulting weights are positive even if some conditional first stages are not. Słoczyński (2026) argues that this makes focusing on  $\beta_{\text{AI}}$  a useful compromise under weak monotonicity.

Focusing on  $\beta_{\text{AI}}$ , however, also raises practical concerns. Because the instrument-covariate interactions can generate many effective instruments relative to the sample size, 2SLS estimation of  $\beta_{\text{AI}}$  may overfit the first stage and suffer from many-instrument bias. Jackknife IV estimators, such as the fixed effect jackknife IV (FEJIV) estimator of Chao et al. (2023) and the unbiased jackknife IV estimator (UJIVE) of Kolesár (2013), are designed to remove this source of own-observation bias. Similar concerns arise in judge leniency designs, where the many instruments typically come from assignment to many decision-makers rather than from instrument-covariate interactions. In that context, Goldsmith-Pinkham et al. (2025) recommend UJIVE and explain why conventional weak instrument diagnostics based on the first-stage  $F$ -statistic are not directly applicable with many instruments. As an alternative, Mikusheva and Sun (2022) propose a pretest for weak identification that assesses whether multiple instruments are jointly strong enough for consistent estimation. Although the pretest is not formally designed for settings with many covariates, the simulations in Słoczyński (2026) suggest that it effectively distinguishes cases in which FEJIV and UJIVE perform well from those in which all estimators of  $\beta_{\text{AI}}$  perform poorly.

Appendix Table A2 replicates several of the estimates of causal effects of pretrial detention on incarceration length in Table 8 in Słoczyński (2026). The specification is the same as in Table 5 above. The linear IV estimate suggests that pretrial detention leads to a large, statistically significant increase in incarceration length of about 666 days. We know, however, that the underlying estimand is not “weakly causal” because of violations of (strong) monotonicity. Indeed, the 2SLS estimate of  $\beta_{\text{AI}}$ , which eliminates the negative weights, suggests a statistically significant increase in incarceration length of about 130 days, which is 80% smaller than the linear IV estimate. While the 2SLS estimate might be more believable, it suffers from many-instrument bias, unlike the associated FEJIV and UJIVE estimates. These, however, suggest an even smaller increase in incarceration length of 51 or 56 days, respectively, and are not statistically different from zero. This conclusion contrasts sharply with the initial IV and 2SLS estimates.<sup>8</sup> It is also supported by the fact that Miku-

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<sup>8</sup>To be clear, Stevenson (2018) does not use either of the problematic estimation approaches as her method of choice. Instead, she uses a nonsaturated specification with a large number of interacted instruments, and estimates the model using the jackknife IV estimator (JIVE) of Angrist et al. (1999). Unlike FEJIV and UJIVE, this estimator is not formally appropriate for settings with many covariates.



sheva and Sun (2022)’s pretest rejects weak identification, as reported by Słoczyński (2026), which indicates that consistent estimation of  $\beta_{AI}$  is possible.

Panel D of Appendix Table A1 lists several MATLAB, R, and Stata packages that can be used to implement FEJIV, UJIVE, and other relevant estimators, as well as the pretest in Mikusheva and Sun (2022).

## Conclusion

The local average treatment effect framework of Angrist and Imbens has fundamentally reshaped how applied microeconomists think about instrumental variables. It is now common to acknowledge that IV models only identify average treatment effects for compliers, examine the characteristics of that subpopulation, and develop careful arguments in favor of instrument validity. In this paper, we have explored three areas in which we believe empirical practice could further benefit from closer engagement with the recent theoretical literature. As we outline above, we recommend that practitioners explicitly target the “true” LATE rather than the usual IV estimand, consider flexible functional forms to avoid parametric misspecification, and use formal tools in response to possible assumption violations, including statistical tests of instrument validity and specifications with many interacted instruments.

## References

- Abadie, Alberto**, “Semiparametric Instrumental Variable Estimation of Treatment Response Models,” *Journal of Econometrics*, 2003, 113 (2), 231–263.
- Ahrens, Achim, Christian B. Hansen, Mark E. Schaffer, and Thomas Wiemann**, “Model Averaging and Double Machine Learning,” *Journal of Applied Econometrics*, 2025, 40 (3), 249–269.
- Andresen, Martin E.**, “montest: Testing LATE Assumptions and Monotonicity Using Machine Learning,” 2026, <https://github.com/martin-andresen/montest>.
- Angrist, Joshua D. and Guido W. Imbens**, “Two-Stage Least Squares Estimation of Average Causal Effects in Models with Variable Treatment Intensity,” *Journal of the American Statistical Association*, 1995, 90 (430), 431–442.
- **and Jörn-Steffen Pischke**, *Mostly Harmless Econometrics: An Empiricist’s Companion*, Princeton–Oxford: Princeton University Press, 2009.
- **, Guido W. Imbens, and Alan B. Krueger**, “Jackknife Instrumental Variables Estimation,” *Journal of Applied Econometrics*, 1999, 14 (1), 57–67.
- **, — , and Donald B. Rubin**, “Identification of Causal Effects Using Instrumental Variables,” *Journal of the American Statistical Association*, 1996, 91 (434), 444–455.
- Autor, David, Andreas Kostøl, Magne Mogstad, and Bradley Setzler**, “Disability Benefits, Consumption Insurance, and Household Labor Supply,” *American Economic Review*, 2019, 109 (7), 2613–2654.

- Balke, Alexander and Judea Pearl**, “Bounds on Treatment Effects from Studies with Imperfect Compliance,” *Journal of the American Statistical Association*, 1997, *92* (439), 1171–1176.
- Belloni, Alexandre, Victor Chernozhukov, Iván Fernández-Val, and Christian Hansen**, “Program Evaluation and Causal Inference with High-Dimensional Data,” *Econometrica*, 2017, *85* (1), 233–298.
- Blandhol, Christine, John Bonney, Magne Mogstad, and Alexander Torgovitsky**, “When Is TSLS Actually LATE?,” *Review of Economic Studies*, 2026, forthcoming.
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel**, “A Practical Guide to Shift-Share Instruments,” *Journal of Economic Perspectives*, 2025, *39* (1), 181–204.
- Carneiro, Pedro, James J. Heckman, and Edward J. Vytlačil**, “Estimating Marginal Returns to Education,” *American Economic Review*, 2011, *101* (6), 2754–2781.
- Carr, Thomas and Toru Kitagawa**, “Testing Instrument Validity with Covariates,” 2023, arXiv:2112.08092.
- Chao, John C., Norman R. Swanson, and Tiemen Woutersen**, “Jackknife Estimation of a Cluster-Sample IV Regression Model with Many Weak Instruments,” *Journal of Econometrics*, 2023, *235* (2), 1747–1769.
- Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins**, “Double/Debiased Machine Learning for Treatment and Structural Parameters,” *Econometrics Journal*, 2018, *21* (1), C1–C68.
- Chyn, Eric, Brigham Frandsen, and Emily Leslie**, “Examiner and Judge Designs in Economics: A Practitioner’s Guide,” *Journal of Economic Literature*, 2025, *63* (2), 401–439.
- Currie, Janet, Henrik Kleven, and Esmée Zwiers**, “Technology and Big Data Are Changing Economics: Mining Text to Track Methods,” *AEA Papers and Proceedings*, 2020, *110*, 42–48.
- Dobbie, Will, Jacob Goldin, and Crystal S. Yang**, “The Effects of Pretrial Detention on Conviction, Future Crime, and Employment: Evidence from Randomly Assigned Judges,” *American Economic Review*, 2018, *108* (2), 201–240.
- Farbmacher, Helmut, Raphael Guber, and Sven Klaassen**, “Instrument Validity Tests with Causal Forests,” *Journal of Business & Economic Statistics*, 2022, *40* (2), 605–614.
- Finkelstein, Amy, Sarah Taubman, Bill Wright, Mira Bernstein, Jonathan Gruber, Joseph P. Newhouse, Heidi Allen, Katherine Baicker, and Oregon Health Study Group**, “The Oregon Health Insurance Experiment: Evidence from the First Year,” *Quarterly Journal of Economics*, 2012, *127* (3), 1057–1106.
- Frölich, Markus**, “Nonparametric IV Estimation of Local Average Treatment Effects with Covariates,” *Journal of Econometrics*, 2007, *139* (1), 35–75.
- Goldsmith-Pinkham, Paul**, “Tracking the Credibility Revolution Across Fields,” 2026, NBER Working Paper no. 35051.
- , **Peter Hull, and Michal Kolesár**, “Leniency Designs: An Operator’s Manual,” 2025, NBER Working Paper no. 34473.
- Heckman, James J. and Edward Vytlačil**, “Structural Equations, Treatment Effects, and Econometric Policy Evaluation,” *Econometrica*, 2005, *73* (3), 669–738.

- Heiler, Phillip**, “Efficient Covariate Balancing for the Local Average Treatment Effect,” *Journal of Business & Economic Statistics*, 2022, 40 (4), 1569–1582.
- Huber, Martin and Giovanni Mellace**, “Testing Instrument Validity for LATE Identification Based on Inequality Moment Constraints,” *Review of Economics and Statistics*, 2015, 97 (2), 398–411.
- Imbens, Guido W. and Donald B. Rubin**, “Estimating Outcome Distributions for Compliers in Instrumental Variables Models,” *Review of Economic Studies*, 1997, 64 (4), 555–574.
- **and Joshua D. Angrist**, “Identification and Estimation of Local Average Treatment Effects,” *Econometrica*, 1994, 62 (2), 467–475.
- Kitagawa, Toru**, “A Test for Instrument Validity,” *Econometrica*, 2015, 83 (5), 2043–2063.
- Kolesár, Michal**, “Estimation in an Instrumental Variables Model with Treatment Effect Heterogeneity,” 2013, unpublished.
- Kwon, Soonwoo and Jonathan Roth**, “Testing Mechanisms,” *Review of Economic Studies*, 2026, forthcoming.
- Ma, Yukun, Pedro H. C. Sant’Anna, Yuya Sasaki, and Takuya Ura**, “Doubly Robust Estimators with Weak Overlap,” 2026, arXiv:2304.08974.
- Maestas, Nicole, Kathleen J. Mullen, and Alexander Strand**, “Does Disability Insurance Receipt Discourage Work? Using Examiner Assignment to Estimate Causal Effects of SSDI Receipt,” *American Economic Review*, 2013, 103 (5), 1797–1829.
- Mao, Minghai, Pedro H. C. Sant’Anna, and Xiaojun Song**, “Falsifying Marginal Treatment Effects,” 2026, unpublished.
- Mikusheva, Anna and Liyang Sun**, “Inference with Many Weak Instruments,” *Review of Economic Studies*, 2022, 89 (5), 2663–2686.
- Mogstad, Magne and Alexander Torgovitsky**, “Identification and Extrapolation of Causal Effects with Instrumental Variables,” *Annual Review of Economics*, 2018, 10, 577–613.
- **and** —, “Instrumental Variables with Unobserved Heterogeneity in Treatment Effects,” in Christian Dustmann and Thomas Lemieux, eds., *Handbook of Labor Economics*, Vol. 5, Amsterdam: Elsevier, 2024, pp. 1–114.
- Mourifié, Ismael and Yuanyuan Wan**, “Testing Local Average Treatment Effect Assumptions,” *Review of Economics and Statistics*, 2017, 99 (2), 305–313.
- Papke, Leslie E. and Jeffrey M. Wooldridge**, “Econometric Methods for Fractional Response Variables with an Application to 401(k) Plan Participation Rates,” *Journal of Applied Econometrics*, 1996, 11 (6), 619–632.
- Poirier, Alexandre and Tymon Słoczyński**, “Quantifying the Internal Validity of Weighted Estimands,” 2025, arXiv:2404.14603.
- Poterba, James M., Steven F. Venti, and David A. Wise**, “401(k) Plans and Tax-Deferred Saving,” in David A. Wise, ed., *Studies in the Economics of Aging*, Chicago–London: University of Chicago Press, 1994, pp. 105–142.
- Ramsey, James Bernard**, “Tests for Specification Errors in Classical Linear Least-Squares Regression Analysis,” *Journal of the Royal Statistical Society: Series B*, 1969, 31 (2), 350–371.
- Sant’Anna, Pedro H. C., Xiaojun Song, and Qi Xu**, “Covariate Distribution Balance via Propensity Scores,” *Journal of Applied Econometrics*, 2022, 37 (6), 1093–1120.

- Singh, Rahul and Liyang Sun**, “Double Robustness for Complier Parameters and a Semi-Parametric Test for Complier Characteristics,” *Econometrics Journal*, 2024, *27* (1), 1–20.
- Śloczyński, Tymon**, “When Should We (Not) Interpret Linear IV Estimands as LATE?,” *Review of Economic Studies*, 2026, forthcoming.
- , **S. Derya Uysal**, and **Jeffrey M. Wooldridge**, “Doubly Robust Estimation of Local Average Treatment Effects Using Inverse Probability Weighted Regression Adjustment,” 2022, arXiv:2208.01300.
- , —, and —, “Abadie’s Kappa and Weighting Estimators of the Local Average Treatment Effect,” *Journal of Business & Economic Statistics*, 2025, *43* (1), 164–177.
- Stevenson, Megan T.**, “Distortion of Justice: How the Inability to Pay Bail Affects Case Outcomes,” *Journal of Law, Economics, and Organization*, 2018, *34* (4), 511–542.
- Sun, Baoluo and Zhiqiang Tan**, “High-Dimensional Model-Assisted Inference for Local Average Treatment Effects with Instrumental Variables,” *Journal of Business & Economic Statistics*, 2022, *40* (4), 1732–1744.
- Sun, Zhenting**, “Instrument Validity for Heterogeneous Causal Effects,” *Journal of Econometrics*, 2023, *237* (2, Part A), 105523.
- Tan, Zhiqiang**, “Regression and Weighting Methods for Causal Inference Using Instrumental Variables,” *Journal of the American Statistical Association*, 2006, *101* (476), 1607–1618.
- Uysal, S. Derya**, “Doubly Robust IV Estimation of the Local Average Treatment Effect,” 2011, unpublished.
- Wooldridge, Jeffrey M.**, *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., Cambridge–London: MIT Press, 2010.
- Zhao, Anqi, Peng Ding, and Fan Li**, “Two-Stage Least Squares with Treatment-Covariate Interactions for Treatment Effect Heterogeneity,” 2026, arXiv:2502.00251.

Table 1: Weights on Covariate-Specific LATEs

| Estimand              | Estimation Procedure                                                                                                                                                                                                                    | $\omega_{m,j}$                                   | What's Overweighted?                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------|
| $\beta_{\text{LATE}}$ | Estimate equation (2) by instrumenting $D_i$ with $Z_i$ ;<br>or estimate $\frac{\mathbb{E}\{\mathbb{E}(Y_i Z_i=1, X_i) - \mathbb{E}(Y_i Z_i=0, X_i)\}}{\mathbb{E}\{\mathbb{E}(D_i Z_i=1, X_i) - \mathbb{E}(D_i Z_i=0, X_i)\}}$ directly | $p_j \pi_j$                                      | Nothing; cells are weighted in proportion to their complier shares |
| $\beta_{\text{IV}}$   | Estimate equation (1) by instrumenting $D_i$ with $Z_i$                                                                                                                                                                                 | $p_j \pi_j \cdot \text{Var}[Z_i   X_{ij} = 1]$   | Cells where assignment varies more                                 |
| $\beta_{\text{AI}}$   | Estimate equation (1) by instrumenting $D_i$ with interactions of $X_i$ and $Z_i$                                                                                                                                                       | $p_j \pi_j^2 \cdot \text{Var}[Z_i   X_{ij} = 1]$ | Cells where the first stage is stronger and assignment varies more |

*Notes:*  $\omega_{m,j}$  denotes the (unnormalized) weight in the expression  $\beta_m = \sum_j \frac{\omega_{m,j}}{\sum_k \omega_{m,k}} \tau_j$  for estimand  $m \in \{\text{LATE}, \text{IV}, \text{AI}\}$ .  $p_j = \mathbb{P}(X_{ij} = 1)$  denotes the share of individuals with  $X_{ij} = 1$ .  $\pi_j = \mathbb{P}(D_i(1) > D_i(0) | X_{ij} = 1)$  denotes the proportion of compliers conditional on  $X_{ij} = 1$ .

Table 2: Causal Effects of Medicaid on the Number of Outpatient Visits

| A. Full Sample                                                |             |                                             |               | $\hat{\beta}_{\text{LATE}}$ | $\hat{\beta}_{\text{IV}}$ | $\hat{\beta}_{\text{AI}}$ |                |
|---------------------------------------------------------------|-------------|---------------------------------------------|---------------|-----------------------------|---------------------------|---------------------------|----------------|
|                                                               |             |                                             |               | 1.001                       | 1.001                     | 1.010                     |                |
|                                                               |             |                                             |               | (0.134)                     | (0.134)                   | (0.132)                   |                |
| B. Three-Person Households                                    |             |                                             |               | $\hat{\beta}_{\text{LATE}}$ | $\hat{\beta}_{\text{IV}}$ | $\hat{\beta}_{\text{AI}}$ |                |
|                                                               |             |                                             |               | 4.022                       | 2.790                     | 2.268                     |                |
|                                                               |             |                                             |               | (1.189)                     | (1.296)                   | (1.118)                   |                |
| C. Estimated Weights in the Sample of Three-Person Households |             |                                             |               |                             |                           |                           |                |
| Survey Wave                                                   | $\hat{p}_j$ | $\widehat{\text{Var}}[Z_i \mid X_{ij} = 1]$ | $\hat{\pi}_j$ | $\hat{w}_{\text{LATE},j}$   | $\hat{w}_{\text{IV},j}$   | $\hat{w}_{\text{AI},j}$   | $\hat{\tau}_j$ |
| 1                                                             | 0.241       | 0.168                                       | 0.455         | 0.351                       | 0.600                     | 0.723                     | 1.067          |
| 2                                                             | 0.448       | 0.037                                       | 0.280         | 0.401                       | 0.151                     | 0.112                     | 6              |
| 3                                                             | 0.310       | 0.099                                       | 0.250         | 0.248                       | 0.249                     | 0.165                     | 5              |

*Notes:* The source of data is Finkelstein et al. (2012). The outcome is the number of outpatient visits in the last six months, as in Table V of Finkelstein et al. (2012). The sample sizes are 23,441 and 58 in Panels A and B, respectively. Standard errors are in parentheses and are clustered at the household level. Panel C is also restricted to the subsample of 58 individuals in three-person households.  $\hat{p}_j$  is the proportion of observations in covariate cell (survey wave)  $j$ .  $\widehat{\text{Var}}[Z_i | X_{ij} = 1]$  is the variance of  $Z_i$  in cell  $j$ .  $\hat{\pi}_j$  is the estimated proportion of compliers in cell  $j$ .  $\hat{w}_{\text{LATE},j}$ ,  $\hat{w}_{\text{IV},j}$ , and  $\hat{w}_{\text{AI},j}$  are the estimated (normalized) weights of cell  $j$  in  $\beta_{\text{LATE}}$ ,  $\beta_{\text{IV}}$ , and  $\beta_{\text{AI}}$ , respectively.  $\hat{\tau}_j$  is the estimated LATE on the number of outpatient visits in cell  $j$ . Each of  $\hat{\beta}_{\text{LATE}}$ ,  $\hat{\beta}_{\text{IV}}$ , and  $\hat{\beta}_{\text{AI}}$ , as reported in Panel B, can be obtained as the dot product of  $\hat{\tau}_j$  and the respective weights, i.e.,  $\hat{\beta}_m = \sum_j \hat{w}_{m,j} \cdot \hat{\tau}_j$  for  $m \in \{\text{LATE}, \text{IV}, \text{AI}\}$ .

Table 3: Causal Effects of 401(k) on IRA Participation

| A. Original Specification | $\hat{\beta}_{IV}$ | $\hat{\beta}_{LATE}$<br>(logit) | $\hat{\beta}_{LATE}$<br>(probit) |
|---------------------------|--------------------|---------------------------------|----------------------------------|
|                           | 0.0274<br>(0.0132) | 0.0165<br>(0.0135)              | 0.0175<br>(0.0136)               |
| RESET $p$ -Value          | 0.000              | 0.000                           | 0.000                            |
| B. Flexible Specification | $\hat{\beta}_{IV}$ | $\hat{\beta}_{LATE}$<br>(logit) | $\hat{\beta}_{LATE}$<br>(probit) |
|                           | 0.0167<br>(0.0132) | 0.0177<br>(0.0128)              | 0.0178<br>(0.0128)               |
| RESET $p$ -Value          | 0.346              | 0.153                           | 0.245                            |

*Notes:* The data are Abadie's (2003) subsample of the 1991 Survey of Income and Program Participation (SIPP). The sample size is 9,275. The outcome is an indicator for participation in IRAs, as in Table 3 of Abadie (2003). The original covariate specification in Panel A includes family income, age, age squared, marital status, and family size. The flexible covariate specification in Panel B includes a quintic in family income, a quintic in age, an indicator for marital status, and indicators for each value of family size.  $\hat{\beta}_{LATE}$  corresponds to the normalized IPW estimator of Uysal (2011) and Słoczyński et al. (2025), with logit or probit instrument propensity scores estimated using maximum likelihood. RESET is based on a quartic in fitted values from each initial model of the instrument propensity score. Heteroskedasticity-robust standard errors are in parentheses.

Table 4: DML Estimates of Causal Effects of 401(k) on IRA Participation

|                 | $\hat{\beta}_{IV}$ | $\hat{\beta}_{LATE}$ |
|-----------------|--------------------|----------------------|
| Lasso           | 0.0165<br>(0.0132) | 0.0209<br>(0.0134)   |
| Elastic Net     | 0.0175<br>(0.0132) | 0.0161<br>(0.0146)   |
| Ridge           | 0.0194<br>(0.0132) | 0.0366<br>(0.0133)   |
| Random Forest   | 0.0215<br>(0.0132) | 0.0230<br>(0.0129)   |
| Regression Tree | 0.0287<br>(0.0134) | 0.0290<br>(0.0130)   |
| XGBoost         | 0.0238<br>(0.0134) | 0.0255<br>(0.0133)   |
| Short-Stacking  | 0.0200<br>(0.0132) | 0.0215<br>(0.0128)   |

*Notes:* The data are Abadie’s (2003) subsample of the 1991 Survey of Income and Program Participation (SIPP). The sample size is 9,275. The outcome is an indicator for participation in IRAs, as in Table 3 of Abadie (2003).  $\hat{\beta}_{IV}$  and  $\hat{\beta}_{LATE}$  are from the partially linear and interactive IV models, estimated via `ddml` (Ahrens et al., 2025) with 3 cross-fitting folds. Propensity scores are trimmed at 0.01. The covariate dictionary follows the flexible specification in Table 3. Short-stacking assigns nonnegative weights to all six machine learning algorithms (Ahrens et al., 2025).

Table 5:  $p$ -Values for Tests of Instrument Validity

|                           | BP       | MW        | FS        |
|---------------------------|----------|-----------|-----------|
| Finkelstein et al. (2012) | 0.99996  | 0.971     | 0.9999998 |
| Abadie (2003)             | 0.997    | 0.9999996 | N/A       |
| Stevenson (2018)          | 0.000102 | 0.0241    | 0.000226  |

*Notes:* The source of data and variable choice for Finkelstein et al. (2012) are the same as in Panel A of Table 2. The source of data and variable choice for Abadie (2003) are the same as in Tables 3 and 4. For Stevenson (2018), the data are a sample of 331,971 arrests in Philadelphia. The outcome is incarceration length, defined as the maximum days of an incarceration sentence. The treatment is pretrial detention. The instrument is whether a given case was heard by the most lenient judge, referred to as “Judge C” in Słoczyński (2026). The covariate specification is saturated in the 17 most common offense types, race and gender of the defendant, and three time periods considered by Stevenson (2018). Groups with fewer than three cases heard by “Judge C” or not heard by “Judge C” are dropped. “BP” refers to the Balke and Pearl conditions. “MW” refers to the Mourifié and Wan conditions. “FS” refers to the requirement that the conditional first stage is nonnegative at all covariate values. All tests are implemented using the `montest` command in R.

Appendix Table A1: Selected Software Packages in MATLAB, R, and Stata

| Package Name                        | Software             | Authors                                                            | Description                                                                                        |
|-------------------------------------|----------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| A. Estimating LATE                  |                      |                                                                    |                                                                                                    |
| <code>causalweight</code>           | R                    | H. Bodory and M. Huber                                             | IPW estimation of the LATE using function <code>lateweight</code>                                  |
| <code>drlate</code>                 | Stata                | T. Słoczyński, S. D. Uysal, and J. M. Wooldridge                   | Regression adjustment, IPW, and doubly robust estimation of the LATE                               |
| <code>kappalate</code>              | Stata                | T. Słoczyński, S. D. Uysal, and J. M. Wooldridge                   | IPW estimation of the LATE                                                                         |
| <code>lateffects</code>             | Stata                | StataCorp                                                          | IPW and doubly robust estimation of the LATE                                                       |
| B. Double Machine Learning          |                      |                                                                    |                                                                                                    |
| <code>ddml</code>                   | R and Stata          | A. Ahrens, C. B. Hansen, M. E. Schaffer, and T. Wiemann            | Double/debiased machine learning estimation of IV and LATE parameters                              |
| <code>DoubleML</code>               | R                    | P. Bach, M. S. Kurz, V. Chernozhukov, M. Spindler, and S. Klaassen | Double/debiased machine learning estimation of IV and LATE parameters                              |
| C. Testing LATE Assumptions         |                      |                                                                    |                                                                                                    |
| <code>LATEtest</code>               | R                    | H. Farbmacher                                                      | Tests of the LATE assumptions based on machine learning                                            |
| <code>montest</code>                | R                    | M. E. Andresen                                                     | Tests of monotonicity and LATE assumptions based on machine learning                               |
| <code>TestMechs</code>              | R                    | S. Kwon and J. Roth                                                | Tests of mediation mechanisms and related testable implications of LATE assumptions                |
| D. Estimation with Many Instruments |                      |                                                                    |                                                                                                    |
| <code>fejiv</code>                  | MATLAB, R, and Stata | Q. Lei and T. Słoczyński                                           | Fixed-effect jackknife IV estimation with many instruments                                         |
| <code>manyiv</code>                 | Stata                | M. Caceres Bravo, P. Goldsmith-Pinkham, P. Hull, and M. Kolesár    | IV estimation and inference with many instruments                                                  |
| <code>manyweakiv</code>             | Stata                | L. Sun                                                             | Pretest for weak identification with many instruments using command <code>manyweakivpretest</code> |



Appendix Table A2: Causal Effects of Pretrial Detention on Incarceration Length

| IV    | 2SLS | FEJIV | UJIVE |
|-------|------|-------|-------|
| 666   | 130  | 51    | 56    |
| (233) | (43) | (91)  | (99)  |

*Notes:* The data are Stevenson’s (2018) sample of 331,971 arrests in Philadelphia. The outcome is incarceration length, defined as the maximum days of an incarceration sentence. The treatment is pre-trial detention. The instrument is whether a given case was heard by the most lenient judge, referred to as “Judge C” in Słoczyński (2026). The covariate specification is saturated in the 17 most common offense types, race and gender of the defendant, and three time periods considered by Stevenson (2018). Groups with fewer than three cases heard by “Judge C” or not heard by “Judge C” are dropped. Standard errors are in parentheses.